Planning for research

Biostatistics and Scientific Research

2-3-2026

Second semester- Third year

Dr.Lama Rafieh

Week 2

Outlines

- Planning for research: research problem, question and hypothesis
- Abstract of journal articles
- Introduction of the article
- Materials and Methods
- Results

Purpose of Study

- The words *purpose* or *goal* usually appear in a purpose statement (e.g., The purpose of this study is..., or, The goal of this study is.....
- A statement of purpose identifies the key study variables and their possible interrelationships

Research Purpose Vs.aims& Objective

Statement of purpose

The researcher’s summary of the overall study goal

Research aims or objectives

The specific missions to be achieved by conducting the study

Research problem

Research problem

A situation or circumstance that requires a solution to be described, explained, or predicted.

Problem statement

A statement displaying the research problem.

Sources of Research Problems

- Experience and clinical fieldwork
- literature
- Social issues
- Theory
- Ideas from external sources

Research question and hypothesis

Research questions

The specific queries the researcher wants to answer in addressing the research problem

Hypothesis

The researcher's predictions about relationships among variables

Research hypothesis

- **Definition**

  A hypothesis is a statement of predicted relationship between the independent and dependent variables under investigation.

- The hypothesis provides the reader with an understanding of the researcher's expectations about the study.

- It should be very specific and limited to the piece of research

Simple Versus Complex Hypothesis

Simple hypothesis

- Expresses a predicted relationship between one independent variable and one dependent variable

Complex hypothesis

- States a predicted relationship between two or more independent variables and/or two or more dependent variables

Directional Versus Non-directional Hypothesis

Directional hypothesis

Predicts the direction of a relationship

Non-directional hypothesis

Predicts the existence of a relationship, not its direction

Research Versus Null Hypothesis

Research hypothesis

States the actual prediction of a relationship

Statistical or null hypothesis

Expresses the absence of a relationship (used only in statistical testing)

Null Hypothesis

- May read as "H_0"
- A null hypothesis predicts that <u>no relationship</u> exists between variables.
- The hypothesis subjected to <u>statistical analysis</u> is the null hypothesis.
- It always assumes no relationship between variables.

Purposes of Hypothesis

- Suggests an answer to the research question
- Provides the reader with an understanding of the researcher's expectations
- Guides the research design
- Dictates the type of statistical analysis to be used

Research problem to hypothesis

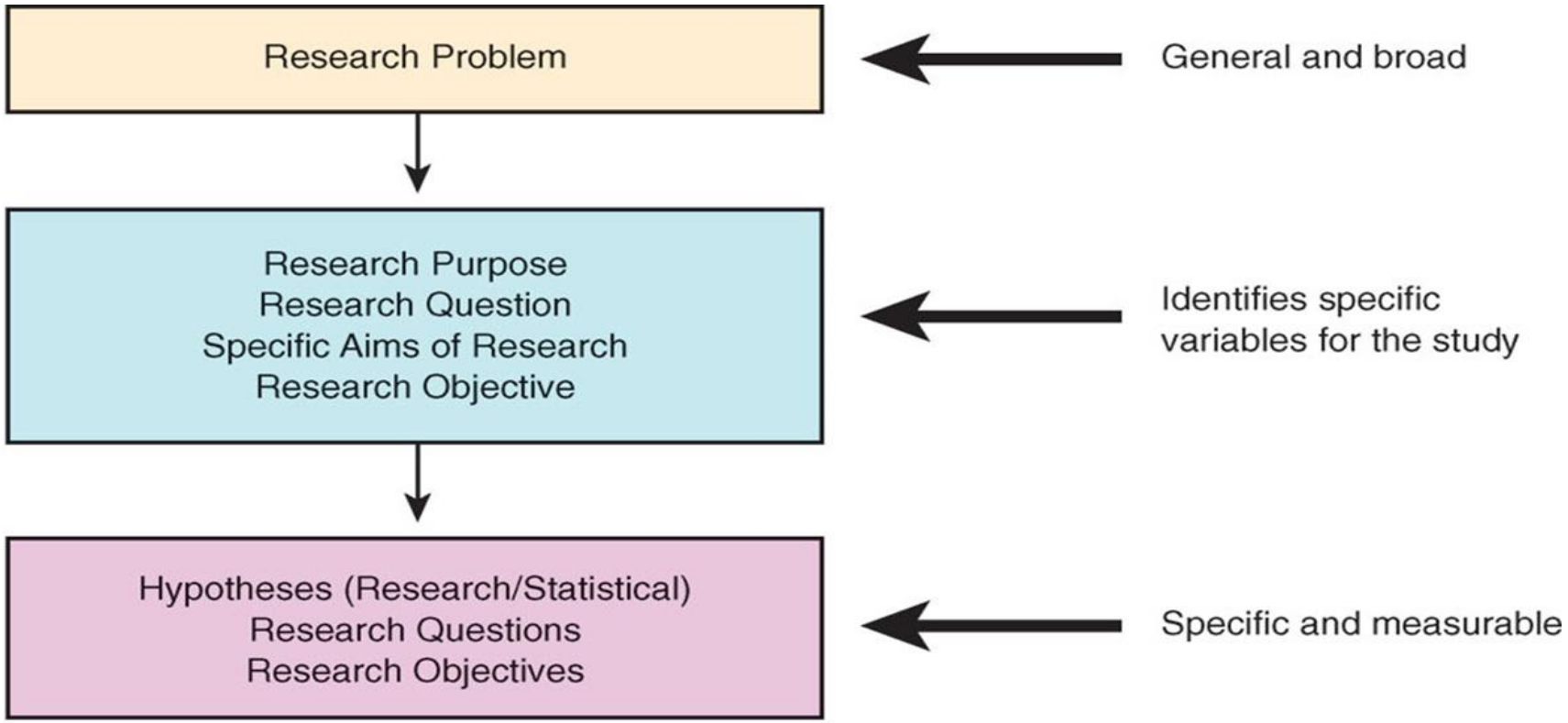

FIGURE 10.3 Levels of development of the statements of the knowledge sought by the study.

Abstract of journal articles

Purpose: Summarize the article's main points, including the research question, methods, results, and conclusions.

Characteristics of a good abstract:

1. Concise: Typically 150-250 words

2. Clear: Easy to understand, even for readers who may not be experts in the field

3. Free of references: Avoid citing specific studies or sources in the abstract

4. Accurate: Reflect the article's content accurately

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Translational Research in Anatomy

journal homepage: www.elsevier.com/locate/tria

Potential teratogenic effect of prenatal dexamethasone administration on palate development: Experimental study in rats

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ARTICLE INFO

Keywords:
Craniofacial development
Palatogenesis
Cleft palate
Dexamethasone
Gestation

ABSTRACT

Background: The interaction of cell populations and synchronization of cell signaling pathways during craniofacial development can cause malformations such as facial clefts when interrupted by teratogenic agents including synthetic corticosteroids. Therefore, this study aimed to determine the potentially disturbing dexamethasone (Dex) effect on palatal shelf development in rat embryos and the possible reasons for this teratogenic potential in relation to fibroblast growth factor 10 (FGF10) as well as FGF10 receptor2 (FGFR2) signaling.
Methods: Thirty pregnant rats were used, which were then randomly categorized into three groups of ten animals each: the control group, where pregnant rats received no treatment, Sham group in which pregnant rats were injected subcutaneously with saline (0.4 mg/kg body weight) during the mid-pregnancy period at gestational day 9–14, and Dex-treated group, where pregnant rats received 0.4 mg/kg of Dex at mid-gestational period. Pregnant animals from all groups were sacrificed on day 20 of gestation before birth and the fetuses were removed for examination of the palatal shelves using a light microscope (LM). In addition, a real-time polymerase chain reaction and a reverse transcription-polymerase chain reaction (RT-PCR) were done to evaluate FGF10 and its receptor FGFR2 gene expression.
Results: The cleft palate incidence rates in the groups of embryos were determined. Only the offspring in the Dex group showed a cleft palate. Moreover, the cleft palate incidence rate in the Dex group was significantly different compared to controls. Moreover, a significant decrease in FGF10 and FGFR2 expression levels was reported in the Dex group than in the controls.
Conclusions: Dex treatment in mid-gestation may increase the incidence of cleft palate development, which may be due to modulation of FGF signaling. This calls for caution when using this medication in the first half of pregnancy unless absolutely necessary and under close medical supervision.

1. Introduction

Antenatal corticosteroids are commonly used to accelerate maturation of fetal lungs in women at risk of preterm birth in order to reduce the incidence of respiratory distress syndrome in premature newborns [1,2]. Steroids are also used in early pregnancy as a treatment for women who experience recurrent miscarriage [3]. Dexamethasone (Dex) intrauterine perfusion is suggested to enhance the clinical outcome of women who suffer from recurrent reproductive failure with abnormally elevated uterine natural killer cells [4]. However, its use during pregnancy is not without potential adverse effects on the embryonic development of the fetus, such as palate formation [5].

The palate is the horizontal separator between the nasal cavity at the top and the oral cavity at the bottom. Anatomically, it consists of two main parts: a hard anterior bony part and a small soft posterior part. The bony part of the palate consists of two palatine plates of the maxilla in front and two horizontal plates of palatine bones in back [6]. While the bony portion is important for feeding and speech, the soft portion is

Abstract of journal articles

Components of an Abstract:

1. Brief background and context;

2. Aim of study;

3. How you will go about the research;

4. The information the research will provide (a brief focused summary of Methods and results);

5. A brief statement of the conclusions drawn from the research.

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Tips for Writing an Abstract

1. Write the abstract last: Write the abstract after completing the rest of the paper.

2. Use clear and concise language to avoid confusing the reader.

3. Avoid using jargon and technical terms unfamiliar to non-experts.

4. Use proper formatting: such as bolding or italicizing, to highlight important information.

5. Proofread carefully to ensure that it is free of errors.

Introduction

Purpose: Provides background information, states the research question or hypothesis, and outlines the study's objectives.

Characteristics of a Good Introduction

1. Clear and concise;

2. Relevant; it should provide relevant background information and context for the research.

3. Focused: The introduction should clearly state the research question or hypothesis and outline the purpose and scope of the study.

4. Easy to understand and free of unnecessary detail;

5. Attractive: it should grab the reader's attention and encourage them to read on.

Introduction

Components:

1. Background information: Provide relevant background information and context for the research.

2. Research question or hypothesis: Clearly state the research question or hypothesis.

3. Purpose and scope: Outline the purpose and scope of the study.

4. Significance: Explain the significance of the research and its potential impact.

5. Thesis statement: Provide a concise thesis statement summarizing the main argument or claim.

Tips for Writing an Introduction

1. Start with a hook: Use an interesting fact, quote, or anecdote to grab the reader's attention.

2. Provide relevant background information and context for the research.

3. Clearly state the research question or hypothesis and outline the purpose & scope of the study.

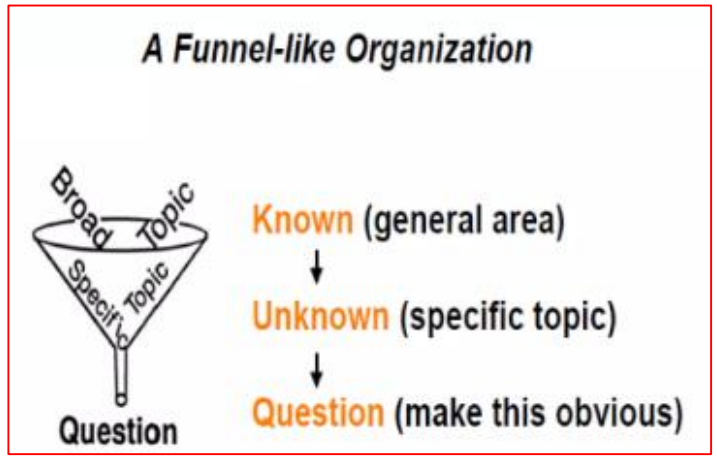
4. Use clear and concise language to avoid confusing the reader.

5. Revise and edit the introduction to ensure it is concise, and engaging.

Tips for Writing an Introduction

- The introduction usually follows a **funnel style**, starting broadly and then narrowing
- Funnel form:

- Known (the considerable picture significance and previous work to
- Unknown (the specific gap in knowledge)
- Question (this should be important, specific and unresolved)



Materials & Methods

Purpose: Describes the research design, procedures, and materials used to collect and analyze the data.

Function:

- Explain clearly how you carried out your study

- This section chronologically describes the process you undertook to complete the research

Character of suitable Methods:

1. Detailed: Provide enough information for the reader to replicate the study;

2. Clear: Easy to understand, even for readers who may not be experts in the field;

3. Avoid unnecessary details;

4. Well-organized: Use a logical structure to present the information.

Materials & Methods

Components:

1. Materials: A description of the materials used in the study, including any equipment, software, or reagents.

2. Methods: A description of the methods used to collect and analyze the data, including any experimental procedures, sampling techniques, or statistical analyses.

3. Study design: A description of the study design, including any experimental or control groups.

4. Sampling strategy: A description of the sampling strategy used to select participants or samples.

5. Data collection methods: A description of the methods used to collect the data, including any surveys, interviews, or observations.

6. Data analysis methods: A description of the methods used to analyze the data, including any statistical software or techniques.

Tips for Writing Materials & Methods

1. Use a logical structure to organize the Materials and Methods section, such as grouping similar materials or methods together;

2. Use clear and concise language to describe the materials and methods used;

3. Avoid unnecessary detail;

4. Use proper citations and credit any sources that provided materials or methods used in the study;

5. Proofread the Materials and Methods section carefully to ensure it is error-frees.

Results

Purpose: Presents the study's findings, including any data visualizations, tables, or figures.

Function: Offers objectively your key results, without interpretation, in an orderly and logical sequence using both text and illustrative materials (Tables and Figures)

Characters:

1. Understandable;

2. Clear and concise: The results section should be free of unnecessary detail.

2. Well-organized: The section should be well-organized and easy to follow.

Results

3. Focused: The results section should focus on the data collected during the study, rather than interpreting the results.

4. Use of visual aids: Visual aids such as tables, figures, and graphs can help present complex data clearly and concisely.

5. Objective: The results section should be objective and free of bias.

Results

Components:

1. Description of the data: A description of the data collected during the study, including the sample size, population, and any data collection methods used.

2. Summary statistics: A summary of the key statistics, such as means, medians, and standard deviations.

3. Data visualizations: The use of visual aids such as tables, figures, and graphs to present complex data clearly and concisely.

4. Results of statistical analyses: Any statistical analyses performed on the data, such as t-tests, ANOVA, or regression analyses.

Tips for Writing Results Section

1. Use a logical structure: to organize the results section, such as presenting the data in the order it was collected.

2. Use clear and concise language: to present the results.

3. Use visual aids such as tables, figures, and graphs to present complex data in a clear and concise manner.

4. Avoid interpreting the results instead focusing on presenting the data in a clear and concise manner.

5. Proofread carefully: to ensure that it is free of errors.

THANK
YOU